[METRIC]
TT-C-530C
June 20, 1995
SUPERSEDING
TT-C-530B
July 20, 1978

FEDERAL SPECIFICATION

COATING COMPOUND, RUST INHIBITIVE, FISH OIL BASE

The General Services Administration has authorized the use of this federal specification, by all federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** This specification covers an air-drying pigmented coating for application as a primer to rusty steel surfaces, after a minimum of surface preparation. It contains a maximum VOC of 420 grams per liter.

2. APPLICABLE DOCUMENTS

2.1 **Government Documents.** The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those in effect on the date of the invitation for bids or request for proposal.

Federal Standards:

FED-STD-123	-	Marking for shipment. (Civil Agencies)
FED-STD-141	-	Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.
FED-STD-313	-	Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

Military Standards

MIL-STD-129	-	Marking for Shipment and Storage. (Military Agencies).

(Copies of specifications and standards required by contractors in connections with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.1 **Other Government publications.**

29 CFR 1910.1200
49 CFR Parts 177-190

(Application for copies should be sent to the Superintendent of Documents, ATTN: New Orders, PO Box 371954, Pittsburgh, PA 15259-7954.)

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: General Services Administration, Paints and Chemicals Commodity Center, Engineering and Commodity Management Division 10FTE, 400 15th St. SW, Auburn, WA. 98001.

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2.2 **Non-Government publications**. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those in effect on the date of the invitation for bids or request for proposal.

American Society for Testing and Materials (ASTM) Standards:

B 117 - Salt Spray (Fog) Testing.
D 50 - Yellow, Orange, Red, and Brown Pigments Containing Iron and Manganese.
D 562 - Consistency of Paints Using the Stormer Viscometer.
D 563 - Phthalic Anhydride Content of Alkyd Resins and Resin Solutions.
D 1210 - Fineness of Dispersion of Pigment-Vehicle Systems, Test for.
D 1475 - Density of Paint, Varnish, Lacquer, and Related Products.
D 1959 - Iodine Value of Drying Oils and Fatty Acids.
D 2369 - Volatile Content of Coatings.
D 2698 - Determination of Pigment Content of Solvent-Type Paints by High Speed Centrifuging
D 3272 - Vacuum Distillation of Solvents from Solvent Base Paints for Analysis.
D 3278 - Flash Point of Liquids by Setaflash Closed Tester.
D 3335 - Low Concentrations of Lead, Cadmium, and Cobalt in Paint By Atomic Absorption Spectroscopy.
D 3457 - Method for Preparation of Methyl Esters from Fatty Acids for Determination of Fatty Acid Composition by Gas-Liquid Chromatography.
D 3624 - Test Method for Low Concentrations of Mercury in Paint by Atomic Absorption Spectroscopy.
D 3718 - Test Method for Low Concentrations of Chromium in Paint by Atomic Absorption Spectroscopy.
D 3721 - Specification for Synthetic Red Iron Oxide Pigment.
D 3725 - Method for Semiquantitative Determination of Fish Oil in Drying Oils and Drying Oil Fatty Acids By Gas-Liquid Chromatography.
D 3924 - Standard Environment for Conditioning and Testing Paint, Varnish, Lacquers, and Related Materials.
E 260 - Practice for Packed Column Gas Chromatography.
G 23 - Light- and Water-Exposure Apparatus (Carbon-Arc Type) for Exposure of Nonmetallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 **Order of precedence**. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3 REQUIREMENTS

3.1 **Materials.** Coatings shall consist of red iron oxide as the essential pigment, ready-mixed with a fish oil modified alkyd resin, extenders, and solvents to meet the requirements of this specification.

3.1.1 **Pigment and extenders.** A red iron oxide pigment conforming to ASTM D 3721 shall be used. Allowable pigment extenders shall be calcium borosilicate, hydrated amorphous silica, non-asbestos talc, or a combination thereof, that meets the requirements of 3.1.3. Addition of thixotropic extenders, that meet the requirements of 3.1.3 are permitted.

3.1.2 **Solvent.** The solvent shall be composed of aliphatic hydrocarbon and meet the requirements of 3.1.3, when tested as specified in 4.3.3. The resultant Volatile Organic Compound (VOC) content shall be not greater than 420 grams per liter.

3.1.3 **Prohibited materials.** The manufacturer shall certify that the product does not contain benzene, toluene, xylene (all isomers), ethylbenzene, ketones, halogenated solvent, hydrolyzable chlorine derivatives, ethylene based glycol ethers and their acetates, nor any carcinogen, as defined in 29 CFR 1910.1200.

3.1.4 **Fish oil.** The fish oil used shall be menhaden, herring, or a combination of both..

3.2 **Quantitative requirements.** The coating shall meet the quantitative requirements specified in table I.

3.3 **Performance requirements.**

3.3.1 **Condition in container.** When tested as specified in table II, a freshly opened full container, shall show no livering, curdling, gelling, caking, or settling that cannot be redispersed with a paddle to a smooth, homogeneous state.

3.3.2 **Appearance of dried film.** When tested as specified in 4.3.7, the brushed and sprayed films shall be smooth and show no wrinkling or blistering.

3.3.3 **Salt spray resistance.** When tested as specified in 4.3.8, there shall be no evidence of rust, blistering, or loss of adhesion on the broad face of the panel, except in the area 5 mm (0.2 inch) from either side of the scored line.

3.3.4 **Accelerated weathering.** When tested as specified in 4.3.9, there shall be no evidence of rust, blistering, or loss of adhesion on the broad face of the panel.

3.3.5 **Storage stability (partially full container).** There shall be no skinning after 72 hours storage, when tested as specified in table II.

3.4 **MSDS DATA.** Material Safety Data Sheets (MSDS) shall be submitted in accordance with FED-STD-313 (see 6.2).

4 QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein, using

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facilities approved by the government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Table I. Quantitative requirements

Characteristic	Minimum	Maximum
Pigment, percent by weight of coating	44	---
Fe_2O_3, percent by weight of pigment 1/	44	---
Nonvolatile vehicle, percent by weight of coating	28	---
Flash point (Setaflash Closed Tester)	38°C (100°F)	---
Viscosity, Krebs units	68	78
Fineness of grind	4	---
Weight per gallon	10.5	---
Iodine number of extracted fatty acids	115	---
Dry time:		
Set-to-touch, hours	---	2
Dust free, hours	---	6
Dry hard, hours	24	48
Sag at 1.5 mil dry film thickness	---	none
Lead content of dry film, percent by weight	---	0.06
Mercury content, ppm	---	10
Chromium content, percent by weight	---	0.005
Cadmium content, percent by weight	---	0.005

1/ This is the minimum iron oxide present, not the minimum amount of iron oxide pigment.

4.1.1 **Responsibility for compliance**. Items shall meet all the requirements of this specification. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 **Rejection**. Failure of a sample to meet any test or inspection requirement specified herein shall be cause for the rejection of the lot from which the sample came. No further lots will be accepted until the supplier has presented sufficient evidence to show that all specification requirements will be met in future lots offered to the government.

4.1.3 **Reduced testing.** When specified by the procuring activity, reduced testing shall be permitted. The procuring activity shall specify which tests are to be performed (see 6.4).

4.2 **Sampling and inspection.**

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4.2.1 **Sampling and inspection for preparation for delivery.** Sampling and inspection shall be in accordance with method 1022 of FED-STD-141.

4.2.2 **Sampling of filled containers.** Samples of filled unit containers, prepared for delivery, shall be taken in accordance with method 1022 of FED-STD-141. The samples shall be inspected for fill and condition in container (see 3.3.1 and table II).

4.2.3 **Sampling for testing.** A four liter (1 gallon) sample of primer shall be randomly selected from the lot and tested in accordance with 4.3.

4.3 **Tests.**

4.3.1 **Testing procedures.** Samples selected in accordance with 4.2.3, shall be tested according to the test methods specified in Table II. Unless otherwise specified, for routine and referee testing all samples shall be prepared and tested at room temperature and the standard atmosphere as defined in ASTM D 3924. All test reports shall contain the individual values utilized in expressing the final result and be evaluated for conformance to the requirements specified in section 3.

4.3.2 **Determination of fish oil.** Separate the vehicle from the pigment using ASTM D 2698. It may be necessary to centrifuge at 40,000 g's for 30 minutes to separate pigment and vehicle. Save the pigment for the test in 4.3.4. Test the vehicle portion of the paint according to ASTM D 563, except the filtrate shall be saved and the precipitate discarded. Test the filtrate according to ASTM D 3725 to verify compliance with 3.4.

4.3.3 **Solvent analysis.**

4.3.3.1 **Solvent extraction.** The solvent shall be extracted from the paint in accordance with ASTM D 3272.

4.3.3.2 **Solvent composition.** The aromatic and oxygenated solvent content shall be determined as specified in 4.3.3.2.1. The presence of halogenated solvents shall be determined as specified in 4.3.3.2.2.

4.3.3.2.1 **Aromatic and oxygenated solvents.** The analysis shall be performed in accordance with method 7356 of FED-STD-141 using a 1.83 m (6 ft.) column. Use phenylcyclohexane as the internal standard. Six-tenths of a milliliter of internal standard shall be added to 3 mL of the distillate. The sample shall be injected in accordance with method 7356. The aliphatic solvents will emerge within one minute and the complete chromatogram should develop in about five minutes. Peaks emerging after one minute are aromatic solvents along with any oxygenated solvents that may be present. Check for compliance with 3.1.2 and 3.1.3. ASTM E 260 will be used if there is questions as to the identity of the chromatographic peaks.

4.3.3.2.2 **Halogenated compounds.** The presence of halogenated compounds shall be determined in accordance with method 5132 of FED-STD-141. Evaluate the results for compliance with 3.1.3.

4.3.4 **Fe_2O_3 content.** Using the pigment obtained in 4.3.2. determine the percent Fe_2O_3 in the total pigment according to ASTM D 50. Check for compliance with the requirement in table I.

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TABLE II Index

Characteristics	Requirement reference	Applicable Test Method		
		FED-STD-141	ASTM Method	Test paragraph reference
Fish oil	3.4	-----	D 3725	4.3.2
Solvent analysis	3.1.2	7356	D 3272	4.3.3
VOC content	3.1.2	-----	D 3960	-----
Pigment content	Table I	----	D 2698	----
Fe_2O_3 content	Table I	----	D 50	4.3.4
Nonvolatile vehicle	Table I	-----	D 2698	----
Flash point	Table I	----	D 3278	-----
Viscosity	Table I	----	D 562	----
Fineness of grind	Table I	-----	D 1210	-----
Weight per gallon	Table I	----	D 1475	----
Iodine number	Table I	----	D 1959	----
Dry time	Table I	2021, 4061	----	4.3.5
Sag	Table I	4494	----	----
Lead content	Table I	----	D 3335	4.3.6
Chromium content	Table I	----	D 3718	-----
Mercury content	Table I	----	D 3624	----
Cadmium content	Table I	----	D 3335	-----
Aromatic and oxygenated solvents	3.1.3	7356	----	4.3.3.2.1
Halogenated compounds	3.1.3	5132	----	4.3.3.2.2
Condition in container	3.3.1	3011	----	----
Appearance of dried film	3.3.2	----	----	4.3.7
Salt spray resistance	3.3.3	----	B 117	4.3.8
Accelerated weathering	3.3.4	----	G 53	4.3.9
Storage stability (partially full container)	3.3.5	3021	----	----

4.3.5 **Drying time.** Using a film applicator, apply the coating to a dry film thickness of 38 µm (0.0015 in) to a window glass panel prepared as specified in method 2021 of FED-STD-141. Determine set-to-touch, dust free, and dry hard times as specified in method 4061 of FED-STD-141. Check for compliance with the requirements in table I.

4.3.6 **Lead content.** Determine lead content in accordance with ASTM D 3335 or by X-ray fluorescence spectroscopy in accordance with the instrument manufacturer's manual. In case of dispute, the X-ray method shall be used.

4.3.7 **Appearance of dried film.**

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4.3.7.1 **Panels.** The panels used shall be window glass conforming to and prepared in accordance with Method 2021 of FED-STD-141.

4.3.7.2 **Brushing.** Apply 2 coats of the coating to the panels, to a total dry film thickness of 76 µm (0.003 in). Allow 24 hours drying time between coats. No top coat shall be applied. Thinner, conforming to TT-T-291, may be added, if necessary; to the primer. However, the resultant VOC shall not be greater than 420 grams per liter. Check for compliance with 3.3.2.

4.3.7.3 **Spraying.** The distance from the test panel to the gun shall be eight to ten inches. Apply 2 coats of the coating to the panels, prepared as specified above, to a total dry film thickness of 76 µm (0.003 in). Allow 24 hours drying time between coats. Thinner, conforming to TT-T-291, may be added, if necessary; to the primer. However, the resultant VOC shall not be greater than 420 grams per liter. Check for compliance with 3.3.2.

4.3.8 **Salt spray resistance.**

4.3.8.1 **Test panel preparation.** Hot-rolled steel test panels shall be descaled in dilute sulfuric acid, and then placed in the salt spray (fog) chamber for 24 hours in accordance with ASTM B 117 in order to acquire a coat of rust. Just prior to use, the panels shall be wire brushed only enough to remove loose rust. Apply 2 coats of the coating to the panels to a total dry film thickness of 76 µm (0.003 in). Allow 24 to 48 hours drying time between coats and 96 hours drying time after the final coat prior to test. A scribed line shall be made on the coated panel as described in ASTM B 117.

4.3.8.2 **Procedure.** The coating shall be tested in the salt spray chamber for 300 hours in accordance with the procedure in ASTM B 117. Check for compliance with 3.3.3..

4.3.9 **Accelerated weathering.** Test panels, prepared as specified in 4.3.8.1, shall be used except that no scribed lines shall be made. Expose the coating for 500 hours in accordance with ASTM Method G 53, using a UV-A-351/302 lamp. The exposure cycle shall be 8 hours of ultraviolet light at 60°C (140°F) followed by 4 hours of condensation at 50°C (122°F) with a constant relative humidity of 50 percent. Check for compliance with 3.3.4.

5 PREPARATION FOR DELIVERY

5.1 **Packaging packing, and marking.** Unless otherwise specified (see 6.2), packaging, packing, and marking shall be in accordance with PPP-P-1892.

6. NOTES.

INFORMATION FOR GUIDANCE ONLY. This section contains information of a general or explanatory nature which is helpful, but is not mandatory.

6.1 **Intended use.** This coating compound is intended for use as a rust-inhibiting primer coating over rusty miscellaneous steel surfaces. However, prior to application, all loose rust, scale or other debris should be removed. Some benefit may result from wiping the metal with solvent. The coating compound should be applied in two coats to provide a total thickness of 3 mils and top-coated with a compatible coating (such as an alkyd-based paint).

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6.2 **Ordering data.** Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Quantity, packaging, packing, and marking required (see 5.1).
(c) Whether reduced testing is allowed and which tests are to be performed (see 4.1.3 and 6.4).
(d) Instructions and address for submission of MSDS (see 3.4).

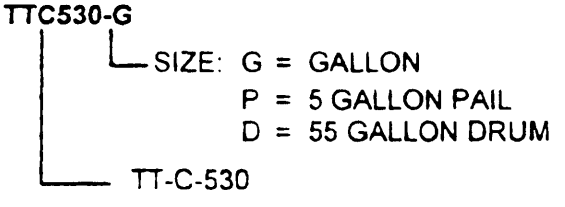
6.3 **Shelf-life surveillance.** For the purpose of shelf-life surveillance, the coating compound stored in the original, unopened container shall show no skinning, livering, curdling, hard, dry caking, or gummy sediment. In addition, the coating shall remix readily to a smooth, homogeneous state, and shall meet the viscosity and dry time requirements in table I.

6.4 **Reduced testing.** When specified (see 4.1.3 and 6.2), the tests for the following requirements should be performed.

a. Condition in container (3.3.1)
b. Flash point (Table I)
c. Viscosity (Table I)
d. Fineness of grind (Table I)
e. Weight per gallon (Table I)
f. Dry time (Table I)

6.5 **Part Identification Number (PIN).** Primer coatings conforming with this Federal Specification shall be identified by a part number configuration consisting of identification of a portion of the specification number and size. An example of the part number configuration is shown below. This part numbering system is intended for identification and cross-indexing of the item within the Federal cataloguing system. Part numbers are not required to be placed on the product or container.

6.5 **Key words.**



Primer
Alkyd
Fish oil
Corrosion preventative

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Low VOC
Iron oxide

MILITARY INTERESTS:

NONE DOD has waived coordination on revisions and amendments to this federal specification until further notice.

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS

Preparing Activity:

GSA-FSS

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